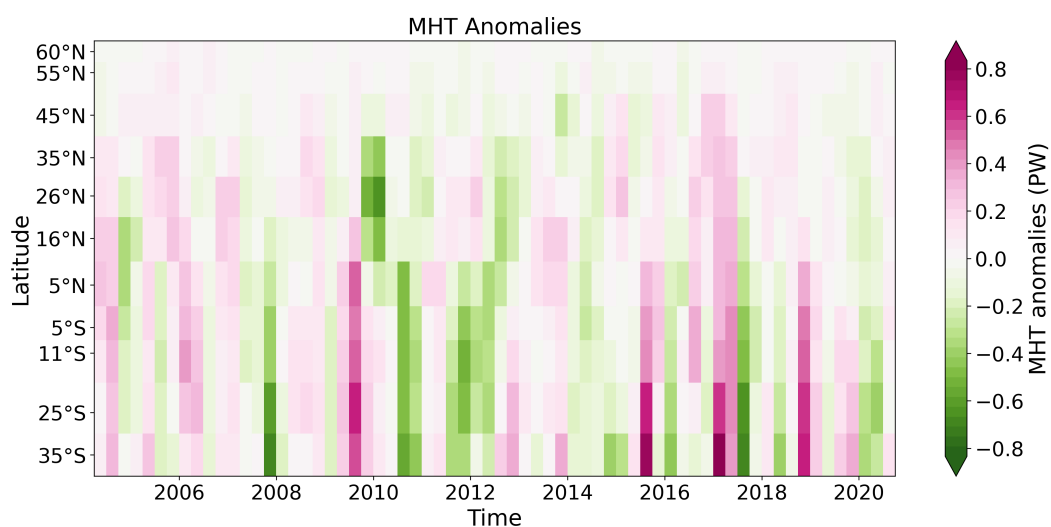


Masterthesis - Plot outline

1.1 How coherent are MHT anomalies?

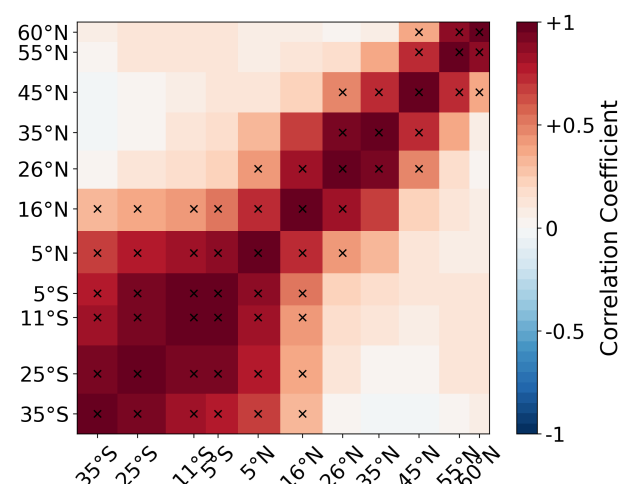
Remark, which I will state in Methods: The following MHT anomalies are processed by deseasonalizing and removing a linear trend. The processed data is used for correlation as well as wavelet analysis. For the connection with NAO, I used only the deseasonalized anomalies and for the heat budget decomposition I evaluated deseasonalized MHT data.

1. First, I am looking at the Hovmöller diagram of the MHT anomalies, to see if overall coherence patterns become visible. The diagram hints towards latitudes where anomalies vary together, namely between 35°N-16°N and 5°N-35°S, where we can see distinct vertical stripes in specific years, which arise in the whole column, sometimes even throughout, as in 2017. Overall the weakest anomalies north of 35°N, in subpolar region.



Hovmöller diagram of detrended, deseasonalized MHT anomalies (PW).

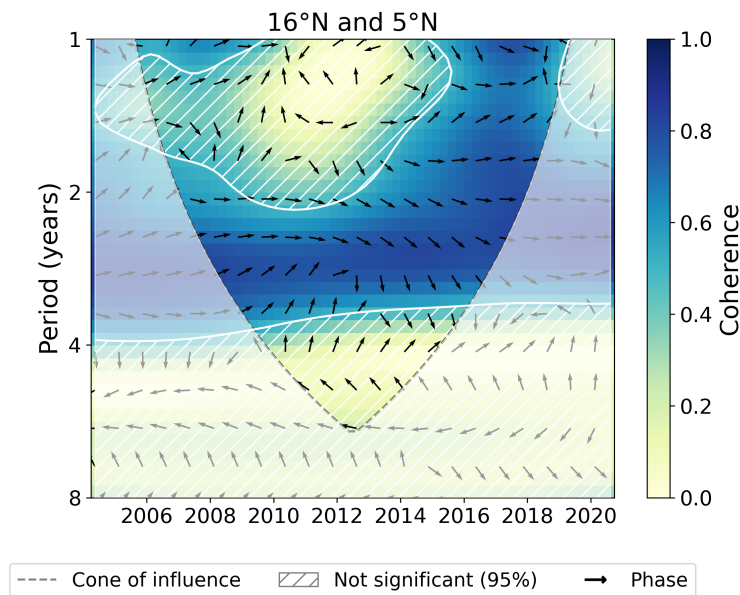
2. To evaluate whether specific pairs or groups of latitudes vary together, I performed a cross-correlation, correlating each latitude with all others at a 0 lag. This revealed that the highest correlations were visible in the groups 35°N-16°N and again in 5°N-35°S. The correlation inside these regions is significant at the 95% level, except between 35°S and 16°N. This suggests that there are two internally coherent regions (35°N-16°N and 5°N-35°S), while the connection between these regions and the area north of 35°N seems less pronounced.



Cross correlation of MHT anomalies at 0 lag.
Significance at 95% level is marked by black crosses.

3. As 16°N - 5°N seems to be the region where the connection between latitudes changes, from internally coherent to disconnected, I will use wavelet coherence to further investigate the coherence of MHT anomalies in this region across frequencies and timescales. Looking at periods between 1 and 2 years the wavelet coherence shows a clear change in pattern. From the outer edge of the cone of influence (COI) the coherence is not significant and low, at around 0.4. Then around 2009, it drops even lower reaching almost 0. This period of low coherence stays persistent until around 2014. Afterwards the coherence rises quickly, reaching significance at the beginning of 2015. Therefore, I distinguish three periods: 2004-2008, 2009-2014 and 2015-2020.

(3.2 for periods around 3 years 16 and 5 stay connected, with some lead lag pattern emerging (2010: 16°N leads 5°N by ~ 0.5 years & then 2014 vice versa (careful here since time period is very short))

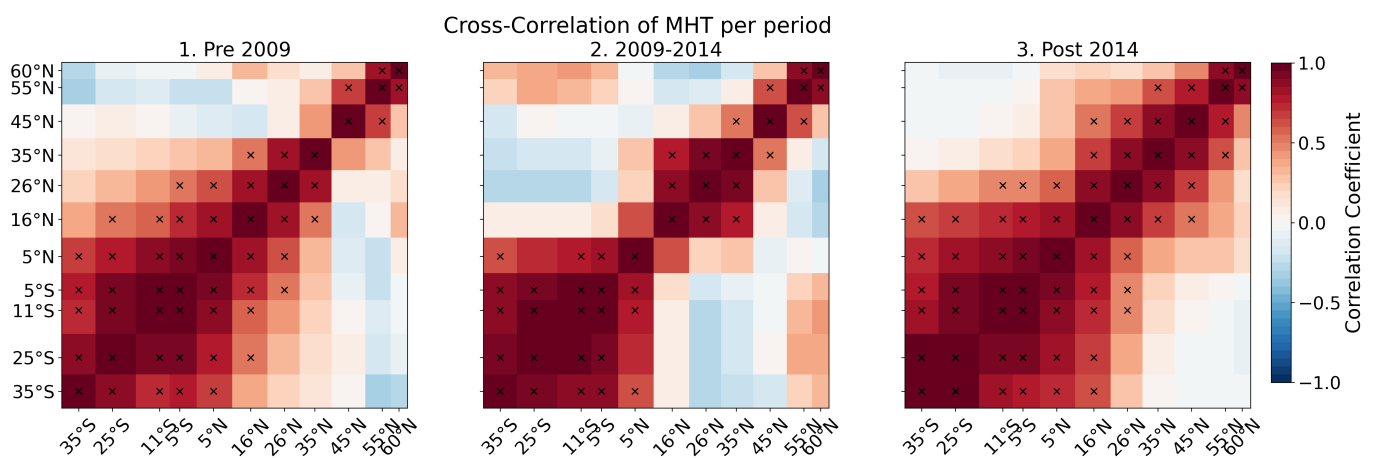


Wavelet coherence analysis between MHT anomalies at 16°N and 5°N using the Morlet wavelet. Areas which are not significant (95% level) are shaded with white stripes and surrounded by white lines. The cone of influence is marked by a dashed grey line and a white shading. Black arrows indicate the phase relationship between the two latitudes, while arrow pointing to the right meaning in phase, pointing left out of phase, pointing upward meaning 16°N leading 5°N by a quarter of the specific period, downward 5°N leading 16°N .

4. MAIN FIGURE

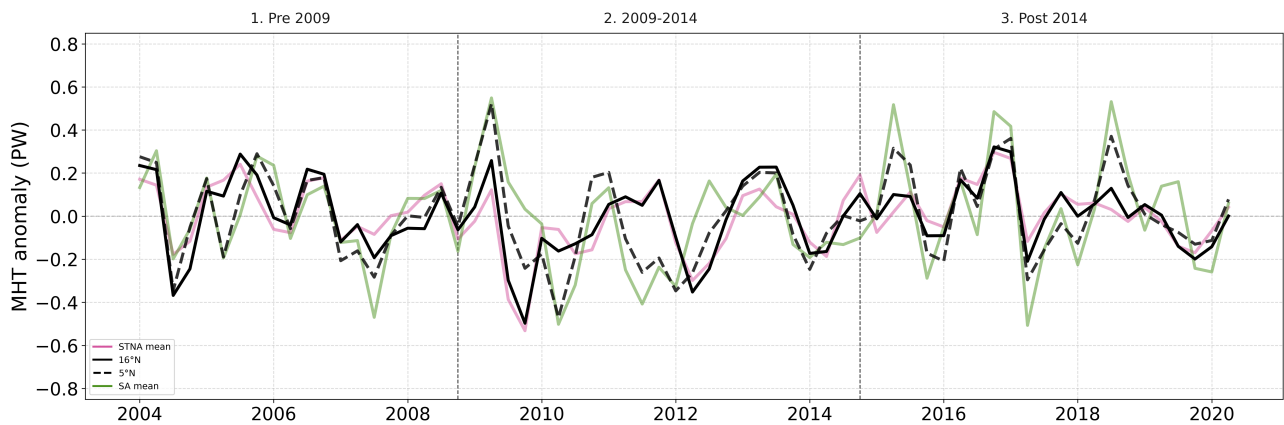
To evaluate how the correlation behaves across all latitudes, I applied the cross correlation analysis of the MHT anomalies for the three time periods (2004-2008, 2009-2014 and 2015-2020) which I identified using the wavelet coherence. This revealed a clear change in pattern: in the first period (2004-2008), the latitudes were strongly correlated across 16°N-5°N. In the second period (2009-2014), the correlation between latitudes of the subtropical North Atlantic (STNA: 35-16°N) and the South Atlantic (SA: 5°N-35°S) drops. This leaves two regions that are internally correlated but disconnected from each other (STNA and SA). In the final period beginning in 2015, the correlation between the two regions increased again, resulting in a stronger connection pattern than in the first period.

Finally, it can be seen that there is a break in the coherence structure across the Atlantic Ocean starting in 2009, with a disconnection between 16°N and 5°N which persists until the end of 2014.

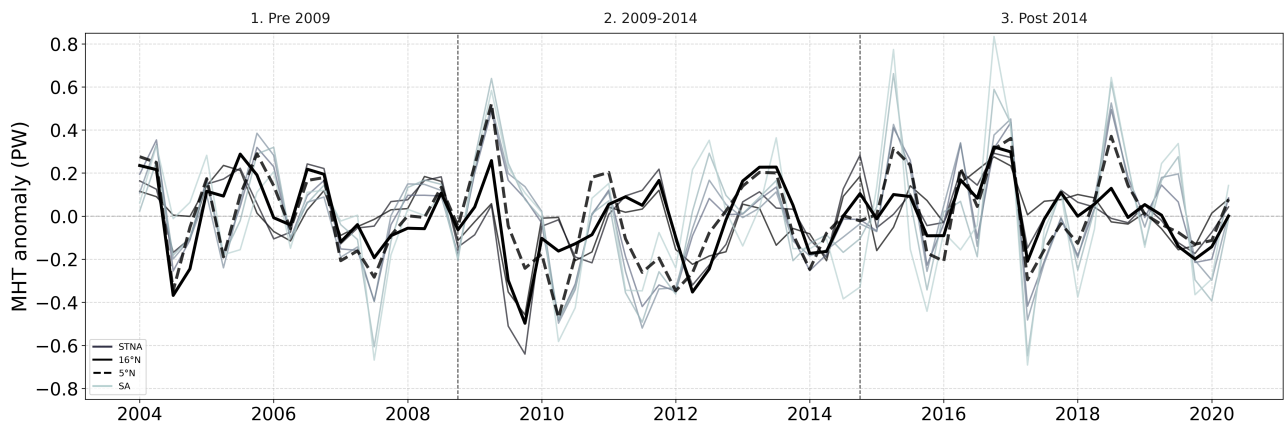


Cross correlation of MHT anomalies at 0 lag, for three time periods: 1. 2004-2008, 2. 2009-2014 and 3. 2015-2020. Significance at 95% level is marked by black crosses.

5. As the correlation lags information about magnitude, I examine the complete timeseries of the MHT anomalies. It became clear that in the first period the STNA and SA are varying together with similar amplitudes, while in the second period this covariability is much less pronounced. For 16°N and 5°N the anomalies are even slightly out of phase in the second period. In the third period the latitudes are varying together again, however the SA mean shows higher overall amplitudes.



Timeseries of MHT anomalies (PW), pink line is the mean of the subtropical north Atlantic latitudes (STNA: 35°N-16°N), while the green line is the mean of the south Atlantic latitudes (SA: 5°N-35°S). The break latitudes are in black, solid black line is 16°N (part of STNA) and dashed black is 5°N (part of SA). The grey dashed vertical lines indicate the different periods.



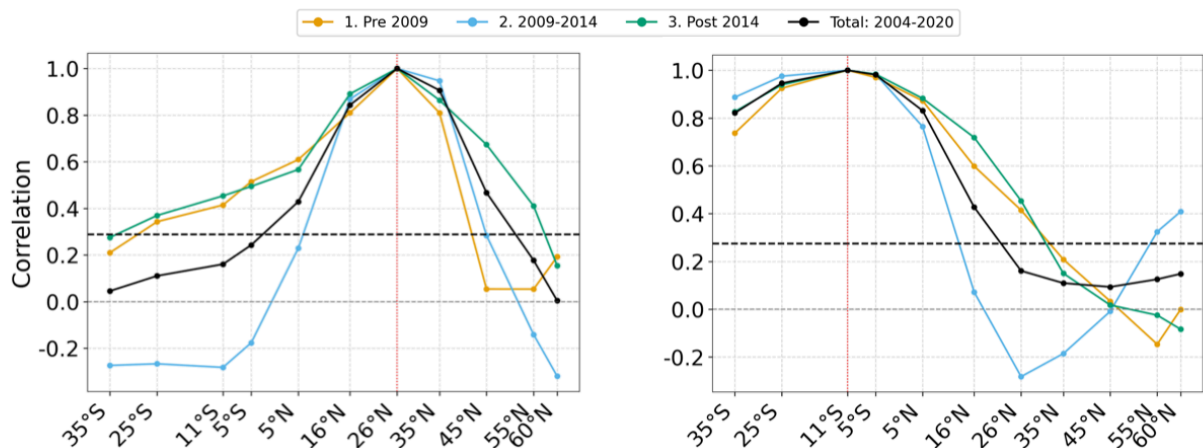
Remark: not sure which plot is best for the timeseries (16 and 5 always highlighted, thicker black lines)

1. Subtropical North Atlantic latitudes (35-16°N) and South Atlantic (5°N-35°S) as mean
2. Individual lines, with dark gray for northern ones and light gray for southern ones

6. The average correlation between the two regions drops for the 2nd period (2009-2014) from 0.53 to a no longer significant value (0.02). After 2014 it increases again to 0.47, while slightly missing the significance threshold of 0.48 (95%).

Remark: if I would use 90% for the significance threshold everywhere, it doesn't change any of the main results, however the correlation reached after 2014 would be significant. Should I still stick to 95%, but showing the threshold of 0.48?

7. The subtropical north Atlantic (STNA) and south Atlantic (SA) blocks remain internally coherent throughout the complete time period (2004-2020).



0-lag correlations between MHT anomalies at 26°N (left) and at 11°S (right) with MHT anomalies at all other latitudes. In color are the different periods, also including the total time period. The black dashed line indicates the 95% significance level for the correlations.

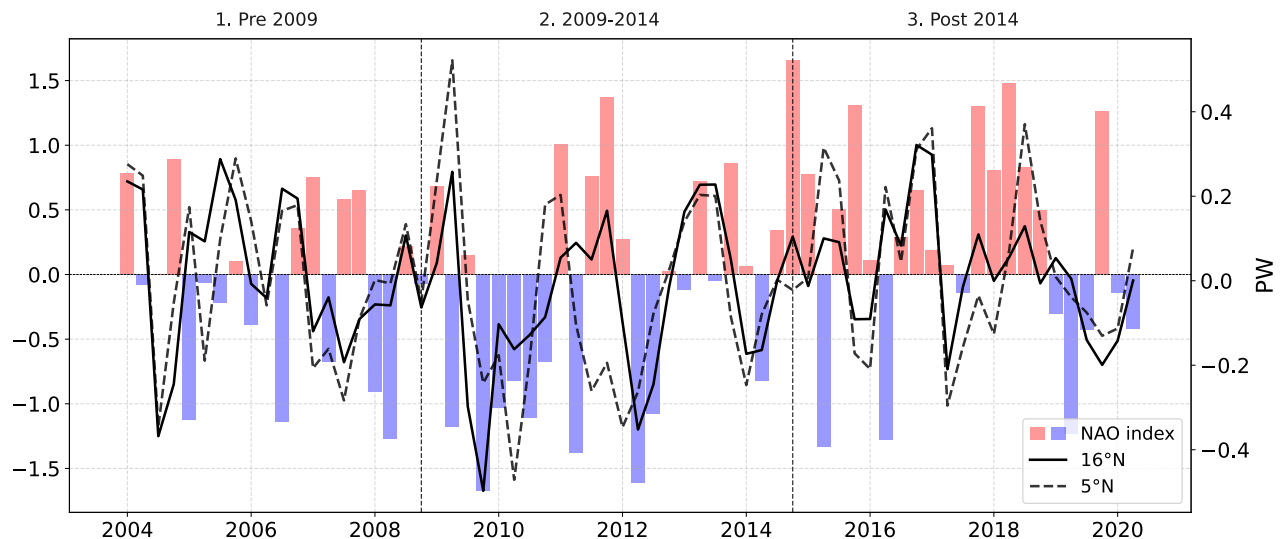
Remark: should I leave the 0-lag figures as a result here or in the appendix?

1.2 Why does the coherence break at 16 and 5?

Now that I established a change in pattern of coherence in a distinct time period (2009-2014) between 16°N and 5°N I will look at possible drivers and changes in the overall heat budget of the region.

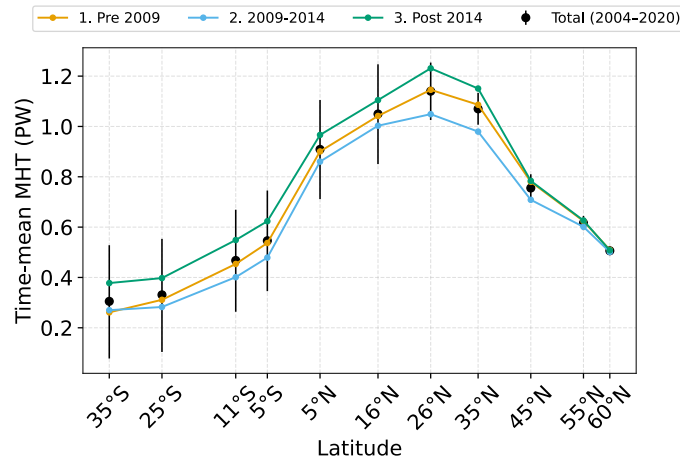
9. To find out if there could be a connection between break of coherence between 16°N and 5°N and the NAO index, I plotted the timeseries for the two MHT anomalies at each latitude against the NAO index and correlated MHT anomalies at 35°N-5°N with it. In the first period (2004-2008) there is no significant correlation between NAO index or MHT anomalies, however in the second period (2009-2014) the correlation increase for the STNA latitudes (35, 26, 16) to around 0.5 (0.52, 0.53, 0.51, significant). However, the correlation with 5°N remains unchanged in this period. This results shows there could be a connection with the anomalous negative NAO index and the MHT anomalies in the period of coherence break.

Remark: Last sentence, not sure if this is enough, or if more interpretation should then go into the discussion?



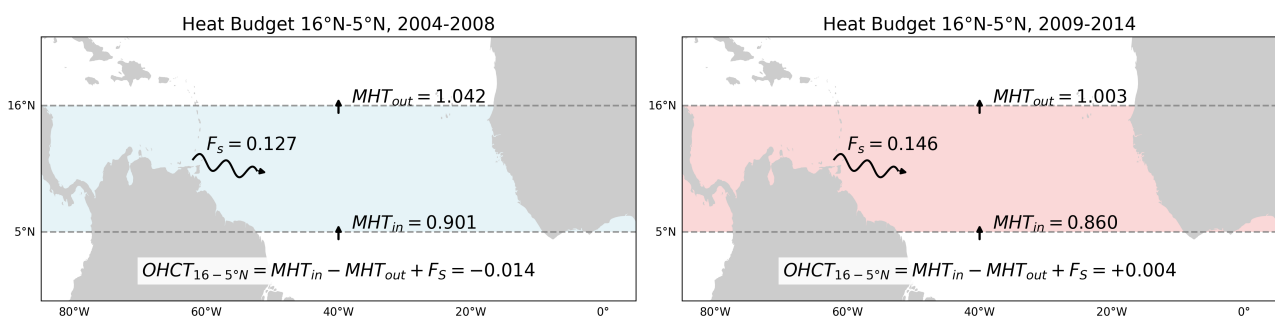
Timeseries of NAO index (left y-axis) and MHT anomalies (right y-axis, in PW) at 16°N (solid black line) and 5°N (dashed black line). NAO index was resampled to quarterly and deseasonalized to align with the MHT anomalies. Positive NAO shaded in red and negative in blue. Vertical dashed lines indicate the three periods identified prior, where the coherence pattern breaks and later rebuilds.

10. When looking at the time-mean heat transport for the different periods as well as the total period, it becomes clear that there is a reduction in MHT across all latitudes in the second period (2009-2014). Afterwards in the period from 2014-2020 the time mean MHT recovers while even surpassing the pre2009 baseline at all latitudes. Concerning the break latitudes, the reduction of MHT at 16°N is around 3.7% while at 5°N around 4.5% in the second period compared to the first period.



Time-mean MHT (PW) for the total timeseries in black, with the 90% confidence interval from the BHM. In color are the different periods.

11. To assess whether the change in coherence can be related to a change in ocean heat content tendency, I decomposed the heat budget into the heat transport contribution and the air sea heat flux contribution in the region between 16°N and 5°N. I found an increase in OHCT, changing from slight cooling to slight warming (delta OHCT = +0.017 PW) which is mainly due to 15% higher heat flux uptake in the region, while the HTC (MHT_in - MHT_out) stays around the same.



Heat budget schematic showing for the region between 16°N and 5°N the ocean heat content tendency (OHCT) and its components MHT_in and MHT_out as well as the surface heat flux F_s. F_s is denoted as positive into the ocean (heat uptake) and negative out (heat release). MHT is defined positive into the region and negative out of the region. When the OHCT the region is shaded blue, as this marks cooling of the region, and red the opposite.

(10 and 11 together show that while there is a reduction of MHT, it is equivalent in amount across latitudes, leading to no change in OHCT by HTC changes, but increase in OHCT by increased heat flux uptake, which is not transported northward. (Discussion: circulation related reduction in MHT, gyre circulation dominantes in states of reduced overturning, 16°N still in subtropical NA gyre, while 5°N not))

Remark: the figure 5 should be my main figure for part 1 of my results which aims to answer the: How coherent is MHT across the atlantic. Then the second part's aim would be to investigate why the break happens at 16 and 5. Not really a main figure for this part?

Appendix

	A	B	C
1	Period	Corr ST & SA block	Significant (95%)
2	Pre 2009	0.53	Yes
3	2009-2014	0.02	No
4	Post 2014	0.47	No (rcrit: 0.48)

